P2 Chapter 11: Integration

Reverse Product Rule (aka By Parts)

SKILL #6: Integration by Parts

$$\int x \cos x \ \ dx = ?$$

Just as the Product Rule was used to **differentiate the product** of two expressions, we can often use 'Integration by Parts' to **integrate a product.**

✏ To integrate by parts:

$$\int u \frac{dv}{dx} dx = uv - \int v \frac{du}{dx} \ dx$$

The Product Rule:

$$\frac{d}{dx}(uv) = v\frac{du}{dx} + u\frac{dv}{dx}$$

On the right-hand-side, both $v\frac{du}{dx}$ and $u\frac{dv}{dx}$ are the product of two expressions. So if we made either the subject, we could use $u\frac{dv}{dx}$ say to represent $x \cos x$ in the example.

Rearranging:

$$u\frac{dv}{dx} = \frac{d}{dx}(uv) - v\frac{du}{dx}$$

Integrating both sides with respect to x, we get the desired formula.

SKILL #6: Integration by Parts

$$\int x \cos x \ \ dx = ?$$

$$\int u \frac{dv}{dx} dx = uv - \int v \frac{du}{dx} \ dx$$

$$u = x \qquad \frac{dv}{dx} = \cos x$$

STEP 1: Decide which thing will be u (and which $\frac{dv}{dx}$).

You're about to differentiate u and integrate $\frac{dv}{dx}$, so the idea is to pick them so differentiating u makes it 'simpler', and $\frac{dv}{dx}$ can be integrated easily.
u will always be the x^n term **UNLESS** one term is $\ln x$.

$$\frac{du}{dx} = 1 \qquad v = \sin x$$

STEP 2: Find $\frac{du}{dx}$ and v.

STEP 3: Use the formula.

$$\int x \cos x \ \ dx = x \sin x - \int 1 \sin x \ \ dx$$
$$= x \sin x + \cos x + C$$

I just remember it as "uv minus the integral of the two new things timesed together"

Another Example

Q Find $\int x^2 \ln x \ dx$

?

Another Example

Q Find $\int x^2 \ln x \ dx$

$u = \ln x$ $\qquad$ $\dfrac{dv}{dx} = x^2$

$\dfrac{du}{dx} = \dfrac{1}{x}$ $\qquad$ $v = \dfrac{1}{3}x^3$

This time, we choose $\ln x$ to be the u because it differentiates nicely.

STEP 1: Decide which thing will be u (and which $\frac{dv}{dx}$).

STEP 2: Find $\frac{du}{dx}$ and v.

STEP 3: Use the formula.

$$\int x^2 \ln x \ dx = \frac{1}{3}x^3 \ln x - \int \frac{1}{3}x^2 \ dx$$

$$= \boldsymbol{\frac{1}{3}x^3 \ln x - \frac{1}{9}x^3 + C}$$

IBP twice! 😐

Q Find $\int x^2 e^x \, dx$

?

IBP twice! 😐

Q Find $\int x^2 e^x \, dx$

$$u = x^2 \qquad \frac{dv}{dx} = e^x$$

$$\frac{du}{dx} = 2x \qquad v = e^x$$

$$\int x^2 e^x \, dx = x^2 e^x - \int 2xe^x \, dx$$

We have to apply IBP again!

$$u = 2x \qquad \frac{dv}{dx} = e^x$$

$$\frac{du}{dx} = 2 \qquad v = e^x$$

$$\int 2xe^x \, dx = 2xe^x - \int 2e^x \, dx$$

$$= 2xe^x - 2e^x$$

Fro Tip: I tend to write out my working for any second integral completely separately, and then put the result back into the original integral later.

Therefore

$$\int x^2 e^x \, dx = x^2 e^x - (2xe^x - 2xe^x) + C$$

$$= x^2 e^x - 2xe^x + 2e^x + C$$

Test Your Understanding

Q Find $\int x^2 \sin x \ dx$

?

Test Your Understanding

Q Find $\int x^2 \sin x \ dx$

$$= 2x \sin x - x^2 \cos x + 2 \cos x + C$$

Integrating $\ln x$ and definite integration

Q Find $\int \ln x \ dx$, leaving your answer in terms of natural logarithms.

?

Q Find $\int_1^2 \ln x \, dx$, leaving your answer in terms of natural logarithms.

?

Integrating $\ln x$ and definite integration

Q Find $\int \ln x \ dx$, leaving your answer in terms of natural logarithms.

$$u = \ln x \qquad \frac{dv}{dx} = 1$$

$$\frac{du}{dx} = \frac{1}{x} \qquad v = x$$

$$\int \ln x \ dx = x \ln x - \int 1 \, dx$$

$$= x \ln x - x + C$$

Q Find $\int_1^2 \ln x \, dx$, leaving your answer in terms of natural logarithms.

$$\int_1^2 \ln x \, dx = [x \ln x - x]_1^2 = \mathbf{2 \ln 2 - 1}$$

If we were doing it from scratch:

$$u = \ln x \qquad \frac{dv}{dx} = 1$$

$$\frac{du}{dx} = \frac{1}{x} \qquad v = x$$

$$\int_1^2 \ln x \, dx = [\boldsymbol{x \ln x}]_{\mathbf{1}}^{\mathbf{2}} - \int_1^2 1 \, dx$$

$$= 2 \ln 2 \ - 1 \ln 1 - [x]_1^2$$

$$= 2 \ln 2 \ - (2 - 1) = 2 \ln 2 - 1$$

In general:

$$\int_a^b u \frac{dv}{dx} dx = [uv]_a^b - \int_a^b v \frac{du}{dx} dx$$

Test Your Understanding

Q Find $\int_0^{\frac{\pi}{2}} x \sin x \ \ dx$

?

Test Your Understanding

Q Find $\int_0^{\frac{\pi}{2}} x \sin x \ dx$

$$u = x \qquad \frac{dv}{dx} = \sin x$$

$$\frac{du}{dx} = 1 \qquad v = -\cos x$$

$$\int x \sin x \ dx = -x \cos x - \int -\cos x \ dx$$

$$= -x \cos x + \int \cos x \ dx$$

$$= -x \cos x + \sin x \ dx$$

$$\therefore \int_0^{\frac{\pi}{2}} x \sin x \ dx = [-x \cos x + \sin x]_0^{\frac{\pi}{2}}$$

$$= \left(-\frac{\pi}{2} \cos \frac{\pi}{2} + \sin \frac{\pi}{2}\right) - (0 + \sin 0)$$

$$= 1$$

One final unusual one…

$$\int e^x \sin x \ dx$$

It doesn't actually matter what you make the u and what the $\frac{dv}{dx}$ this time. But the hard part is realising how to 'close the loop' at the end…
Exam Note: This came up in an exam once and caught an awful lot of students (and teachers!) by surprise.

?

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$$u = e^x \quad \frac{dv}{dx} = \sin x$$

$$\frac{du}{dc} = e^x \quad v = -\cos x$$

$$\int e^x \sin x \ dx = -e^x \cos x - \int -e^x \cos x \ dx$$

$$= -e^x \cos x + \int e^x \cos x \ dx$$

Consider $\int e^x \cos x \ dx$

$$u = e^x \quad \frac{dv}{dx} = \cos x$$

$$\frac{du}{dx} = e^x \quad v = \sin x$$

$$\therefore \int e^x \sin x \ dx = -e^x \cos x + e^x \sin x - \int e^x \sin x \ dx$$

$$2\int e^x \sin x \ dx = -e^x \cos x + e^x \sin x + c$$

This is the surprising bit! Because we ended up with the original expression, we can 'collect' these integrals together!

$$\int e^x \sin x \ dx = -\frac{1}{2} e^x \cos x + \frac{1}{2} e^x \sin x + c'$$

Exercise 11.6

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Extension

$$\int \arcsin(x)\, dx$$

$$= \boxed{\quad ? \quad}$$

(hint: $\frac{dv}{dx} = 1$)

[STEP I 2014 Q2]

(i) Show that $\int \ln(2-x)\,\mathrm{d}x = -(2-x)\ln(2-x) + (2-x) + c$, where $x < 2$.

(ii) Sketch the curve A given by $y = \ln|x^2 - 4|$.

(iii) Show that the area of the finite region enclosed by the positive x-axis, the y-axis and the curve A is $4\ln(2+\sqrt{3}) - 2\sqrt{3}$.

(iv) The curve B is given by $y = \left|\ln|x^2 - 4|\right|$. Find the area between the curve B and the x-axis with $|x| < 2$.

[*Note: you may assume that* $t \ln t \to 0$ *as* $t \to 0$.]

You will need the following standard results (given in your formula booklet) for the main exercise. We'll prove them later. ✐

$$\int \tan x \; dx = \ln|\sec x| + C \qquad \int \sec x \; dx = \ln|\sec x + \tan x| + C$$

$$\int \cot x \; dx = \ln|\sin x| + C \qquad \int \operatorname{cosec} x \; dx = \ln|\operatorname{cosec} x + \cot x| + C$$

Exercise 11.6

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Page 94

Extension

$$\int \arcsin(x)\, dx$$

$$= \sqrt{1 - x^2} + x \arcsin(x) + C$$

(hint: $\frac{dv}{dx} = 1$)

[STEP I 2014 Q2]

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$$\int \cot x \; dx = \ln|sin\, x| + C \qquad \int \operatorname{cosec} x \; dx = \ln|\operatorname{cosec} x + \cot x| + C$$

Homework Exercise

1 Find the following integrals.

a $\int x \sin x \, dx$ **b** $\int x e^x \, dx$ **c** $\int x \sec^2 x \, dx$

d $\int x \sec x \tan x \, dx$ **e** $\int \frac{x}{\sin^2 x} dx$

2 Find the following integrals.

a $\int 3 \ln x \, dx$ **b** $\int x \ln x \, dx$ **c** $\int \frac{\ln x}{x^3} dx$

d $\int (\ln x)^2 \, dx$ **e** $\int (x^2 + 1) \ln x \, dx$

Hint You will need to use these standard results. In your exam they will be given in the formulae booklet:

- $\int \tan x \, dx = \ln|\sec x| + c$
- $\int \sec x \, dx = \ln|\sec x + \tan x| + c$
- $\int \cot x \, dx = \ln|\sin x| + c$
- $\int \operatorname{cosec} x \, dx = -\ln|\operatorname{cosec} x + \cot x| + c$

3 Find the following integrals.

a $\int x^2 e^{-x} \, dx$ **b** $\int x^2 \cos x \, dx$ **c** $\int 12x^2(3 + 2x)^5 \, dx$ **d** $\int 2x^2 \sin 2x \, dx$ **e** $\int 2x^2 \sec^2 x \tan x \, dx$

4 Evaluate the following:

a $\int_0^{\ln 2} x e^{2x} \, dx$ **b** $\int_0^{\frac{\pi}{2}} x \sin x \, dx$ **c** $\int_0^{\frac{\pi}{2}} x \cos x \, dx$ **d** $\int_1^2 \frac{\ln x}{x^2} dx$

e $\int_0^1 4x(1 + x)^3 \, dx$ **f** $\int_0^{\pi} x \cos \frac{1}{4} x \, dx$ **g** $\int_0^{\frac{\pi}{3}} \sin x \ln(\sec x) \, dx$

5 **a** Use integration by parts to find $\int x \cos 4x \, dx$. **(3 marks)**

b Use your answer to part **a** to find $\int x^2 \sin 4x \, dx$. **(3 marks)**

Homework Exercise

6 **a** Find $\int \sqrt{8-x}\,\mathrm{d}x$. **(2 marks)**

b Using integration by parts, or otherwise, show that

$$\int (x-2)\sqrt{8-x}\,\mathrm{d}x = -\tfrac{2}{5}(8-x)^{\frac{3}{2}}(x+2) + c$$ **(6 marks)**

c Hence find $\int_4^7 (x-2)\sqrt{8-x}\,\mathrm{d}x$. **(2 marks)**

7 **a** Find $\int \sec^2 3x\,\mathrm{d}x$. **(3 marks)**

b Using integration by parts, or otherwise, find $\int x\sec^2 3x\,\mathrm{d}x$. **(6 marks)**

c Hence show that $\int_{\frac{\pi}{18}}^{\frac{\pi}{9}} x\sec^2 3x\,\mathrm{d}x = p\pi - q\ln 3$, finding the exact values of the constants p and q. **(4 marks)**

Homework Answers

1 **a** $-x\cos x + \sin x + c$ **b** $x\,e^x - e^x + c$
c $x\tan x - \ln|\sec x| + c$ **d** $x\sec x - \ln|\sec x + \tan x| + c$
e $-x\cot x + \ln|\sin x| + c$

2 **a** $3x\ln x - 3x + c$ **b** $\frac{x^2}{2}\ln x - \frac{x^2}{4} + c$

c $-\frac{\ln x}{2x^2} - \frac{1}{4x^2} + c$ **d** $x(\ln x)^2 - 2x\ln x + 2x + c$

e $\frac{x^3}{3}\ln x - \frac{x^3}{9} + x\ln x - x + c$

3 **a** $-e^{-x}x^2 - 2x\,e^{-x} - 2e^{-x} + c$
b $x^2\sin x + 2x\cos x - 2\sin x + c$

c $x^2(3+2x)^6 - \frac{x(3+2x)^7}{7} + \frac{(3+2x)^8}{112} + c$

d $-x^2\cos 2x + x\sin 2x + \frac{1}{2}\cos 2x + c$
e $x^2\sec^2 x - 2x\tan x + 2\ln|\sec x| + c$

4 **a** $2\ln 2 - \frac{3}{4}$ **b** 1 **c** $\frac{\pi}{2} - 1$
d $\frac{1}{2}(1 - \ln 2)$ **e** 9.8 **f** $2\sqrt{2}\pi + 8\sqrt{2} - 16$
g $\frac{1}{2}(1 - \ln 2)$

5 **a** $\frac{1}{16}(4x\sin 4x + \cos 4x) + c$
b $\frac{1}{32}((1 - 8x^2)\cos 4x + 4x\sin 4x)) + c$

6 **a** $-\frac{2}{3}(8-x)^{\frac{3}{2}} + c$

b $u = x - 2 \Rightarrow \frac{du}{dx} = 1;\ \frac{dv}{dx} = \sqrt{8-x} \Rightarrow v = -\frac{2}{3}(8-x)^{\frac{3}{2}}$

$$I = (x-2)\left(-\frac{2}{3}(8-x)^{\frac{3}{2}}\right) - \int -\frac{2}{3}(8-x)^{\frac{3}{2}}\,dx$$

$$= -\frac{2}{3}(x-2)(8-x)^{\frac{3}{2}} + \frac{2}{3}\int(8-x)^{\frac{3}{2}}\,dx$$

$$= -\frac{2}{3}(x-2)(8-x)^{\frac{3}{2}} - \frac{4}{15}(8-x)^{\frac{5}{2}} + c$$

$$= -\frac{2}{3}(x-2)(8-x)^{\frac{3}{2}} - \frac{4}{15}(8-x)(8-x)^{\frac{3}{2}} + c$$

$$= (8-x)^{\frac{3}{2}}\left(-\frac{2}{3}(x-2) - \frac{4}{15}(8-x)\right) + c$$

$$= (8-x)^{\frac{3}{2}}\left(-\frac{2x}{5} - \frac{4}{5}\right) + c = -\frac{2}{5}(8-x)^{\frac{3}{2}}(x+2) + c$$

c 15.6

7 **a** $\frac{1}{3}\tan 3x + c$ **b** $\frac{1}{3}x\tan 3x - \frac{1}{9}\ln|\sec 3x| + c$

c $\int_{\frac{\pi}{18}}^{\frac{\pi}{9}} x\sec^2 3x = \left[\frac{1}{3}x\tan 3x - \frac{1}{9}\ln|\sec 3x|\right]_{\frac{\pi}{18}}^{\frac{\pi}{9}}$

$$= \left(\frac{\sqrt{3}\pi}{27} - \frac{1}{9}\ln 2\right) - \left(\frac{\sqrt{3}\pi}{162} - \frac{1}{9}\ln\frac{2}{\sqrt{3}}\right)$$

$$= \frac{5\sqrt{3}\pi}{162} - \frac{1}{9}\ln 2 + \frac{1}{9}\ln 2 - \frac{1}{9}\ln\sqrt{3}$$

$$= \frac{5\sqrt{3}\pi}{162} - \frac{1}{18}\ln 3 \Rightarrow p = \frac{5\sqrt{3}}{162} \text{ and } q = \frac{1}{18}$$